Tribhuvan University
Institute of Science and Technology
SCHOOL OF MATHEMATICAL SCIENCES
First Assessment 2081

Subject: **Regression Analysis**
Course No: **MDS 605**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. Explain cross sectional data with suitable example.
2. Show that mean of error/residual is always zero.
3. How do you measure the goodness of fit in multiple regression analysis. Discuss two model selection criteria briefly.
4. Describe the null and alternative hypothesis and state the test statistic and decision criteria to test the significance of individual regression coefficient (βi).
5. Suppose VIF for three independent variables X1, X2, and X3 are 3.18, 4.21, and 13.18 respectively then what decision do you make for the multicollinearity?

Group B [5×6= 30]

6. Describe the general regression model and also describe the assumptions of the regression model.

OR

Show that $\underline{\hat{\beta}} = (X^t X)^{-1} (X^t \underline{Y})$

7. A state fisheries commission wants to estimate the number of bass caught in a given lake during a season in order to restock the lake with the appropriate number of young fish. The commission could get a fairly accurate assessment of the seasonal catch by extensive "netting sweeps" of the lake before and after a season, but this technique is much too expensive to be done routinely. Therefore, the commission samples a number of lakes and records Y (the seasonal catch, thousands of bass per square mile of lake area), X1 (the number of lakeshore residences per square mile of lake area), X2 (the size of lake in square miles), X3 (X3=1, if the lake has public access, 0 if not) and X4 (a structure index. (Structures are weed beds sunken trees, drop-offs and other living places for bass.) The commissions run the data in SPSS to fit the regression model and the incomplete output is obtained as follows:

Predictor	Unstandardized Coefficient		t-value	p-value
	Coefficient	Standard error		
Constant	-1.9378	0.9081	?	
Residence (X1)	0.0193	0.0101	?	0.560
Size (X2)	0.3323	0.2458	?	0.120
Access (X3)	0.8355	0.2250	?	0.001
Structure (X4)	0.0477	0.0050	?	0.001

Analysis of variance (ANOVA) table:

Source	Df	SS	MSS	F	p-value
Regression	?	21.04	?	?	0.000
Error	?	2.82	?		
Total	19	23.86			

a) Complete above coefficient and ANOVA table and describe the best-fitting regression equation for these data.
b) Test the significant impact of X3 (Access) on Y (the seasonal catch, thousands of bass per square mile of lake area) at 5% level of significance.
c) Compute the standard error of the estimate and interpret its meaning.
d) What percentage of the variation in gross earnings is explained by three variables?
e) At the 0.05 level of significance, is the regression model significant as a whole?

8. State the unbiased ness property of estimator of slope/regression coefficient.
9. Describe Breusch-Godfrey Lagrange Multiplier Test (LM test) to test the autocorrelation.

OR

Explain the consequences of autocorrelation on unbisedness, bestness and consistent properties of OLS.

10. Explain Breusch Pagan Godfrey test to examine the presence of heteroskedasticity in data.

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Tribhuvan University
Institute of Science and Technology

SCHOOL OF MATHEMATICAL SCIENCES

Second Assessment 2082

Subject: **Regression Analysis** *Full marks:* **45**
Course No: **MDS 605** *Pass Marks:* **22.5**
Level: **MDS / II Year / III Semester** *Time:* **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. Describe BLUE properties briefly.
2. What is multicollinearity? Explain variance inflation factor criteria briefly.
3. Describe forward selection procedure briefly.
4. Explain techniques for checking normality.
5. How do you measure the goodness of fit in logistic regression.

Group B [5×6= 30]

6. Describe the general regression model and also describe the assumptions of the regression model.

OR

A sample of 34 stores in a chain is selected for a test-market study of Pokhara Noodles. All the stores selected have approximately the same monthly sales volume. Measurements are obtained by taking price and promotional expenditure as independent variables and sales as a dependent variable. The partial output obtained from SPSS Software is as following.

ANOVA

Model	Sum of square	df	Mean square	F	p-Value
Regression	39472731	?	?	?	0.000
Residual	?	?	?		
Total	52093678	?			

Coefficients

Model	Unstandardized Coefficient		T	p-value
	Bi	Standard error		
Constant	5837.521	628.150	?	
Price	-53.217	6.852	?	0.000
Promotion	3.613	0.685	?	0.000

a. Test the significance effect of promotion on sales at 1% level of significance.
b. What percentage of total variation in sales is explained by price and promotional expenditure?
c. Compute adjusted coefficient of determination and interpret its meaning.
d. Compute standard error of estimate and interpret its meaning.
e. Test the overall fit of the model at 5% level of significance.
f. Develop the regression model and predict the sales when price Rs.70 and promotional expenditure is Rs.300.

7. Suppose the residuals for a set of data collected over 10 consecutive time period are as follows. Let the model consists of two independent variables

Time	1	2	3	4	5	6	7	8	9	10
Residuals	-3	-4	-6	-2	-1	0	2	3	4	7

Compute the Durbin Watson test statistic and test for significance at .level 05. Is there evidence of positive autocorrelation among residuals?

8. Describe Breusch-Godfrey Lagrange Multiplier Test (LM test) to test the autocorrelation.
9. Explain the impact of Heteroskedasticity on BLUE property. Explain Breusch-Pagan-Godfrey Test briefly.
10. When do you use the logistic regression? Explain logit model briefly.

OR

Define dummy variable. How does interchanging the code numbers affect the y-intercept and regression coefficient (slope) when we consider dummy variable as an independent variable.

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Exam Roll No..........

Tribhuvan University

Institute of Science and Technology

2082

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MDS / II Year / Third Semester/ Science

Data Science (MDS 605)

(Regression Analysis)

Full Marks: 45

Pass Marks: 22.5

Time: 2 hours.

Candidates are required to give their answers in their own words as for as practicable.
The figures in the margin indicate full marks. The symbols have their usual meanings.
Attempt all questions.

Group A [5 × 3 = 15]

1. Show that var ($\underline{€}$) = $\sigma^2 I$

2. Explain R^2 and Adjusted R^2 with examples.

3. Explain the limitations of Durbin Watson test to test the autocorrelation.

4. Explain dummy variable. If a regression equation consists of following information $\hat{Y} = 20000 - 3000X$, where Y = Salary and X = Gender (Female =0 and Male =1). Interpret the meaning of Y-intercept and slope. What are the expected salaries of male and female?

5. Describe forward selection procedure briefly.

Group B [5 × 6 = 30]

6. Explain assumptions of multiple regression model and BLUE properties.

OR

Multiple regressions are used by accountants in cost analysis to shed light on the factors that cause costs to be incurred and the magnitudes of their effects. The independent variables of such a regression model are the factors believed to be related to cost, the dependent variable. In some instances, however, it is desirable to use physical units instead of cost as the dependent variable in a cost analysis. This would be the case if most of the cost associated with the activity of interest is a function of some physical unit, such as hours of labor. The advantage of this approach is that the regression model will provide estimates of the number of labor hours required under different circumstances and these hours can then be costed at the current labor rate (Horngren, Foster. and Datar, *1994).* The sample data have been collected from a firm's accounting and production records to provide cost information about the firm's shipping department.

The SPSS computer printout for fitting the model $Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + e$ is provided as follows:

Coefficient table:

Predictor	Unstandardized Coefficients (bi)	Sbi	t	P-value
(Constant)	131.92	25.69	?	
X_1	2.73	2.27	?	0.125
X_2	0.05	0.09	?	0.231
X_3	-2.58	0.64	?	0.001

Where Y = Labor (in Hrs)

X_1= Pounds Shipped (in 1000)

X_2= Percentage of Units Shipped by Truck

X_3= Average Shipment Weight

a. Complete the above coefficient table.

b. What is the best-fitting regression equation for these data? What number of labor hours would you expect for 33,000 pounds shipped, 90 percentages of units shipped by truck and 20 average shipment weights.

c. Test the significance of average shipment weight on number of labor hours required.

7. Explain the impact of multicollinearity on BLUE properties. Also describe VIF method to detect the multicollinearity.

8. Describe Breusch-Godfrey Lagrange Multiplier Test (LM test) to test the autocorrelation.

OR

Suppose the residuals for a set of data collected over 10 consecutive time period are as follows. Let the model consists of two independent variables

Time	1	2	3	4	5	6	7	8	9	10
Residuals	-5	-4	-3	-2	-1	1	2	3	4	5

Compute the Durbin Watson test statistic and test for significance at. level 05. Is there evidence of positive autocorrelation among residuals?

9. State the sources of heteroskedasticity. Explain the Goldfeld-Quandt test to test the heteroskedasticity.

10. Explain logit model, odds ratio and wald test in logistic regression briefly.

Tribhuvan University
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SCHOOL OF MATHEMATICAL SCIENCES
First Assessment 2082

Subject: **Regression Analysis**
Course No: **MDS 605**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A **[5x3=15]**

1. Explain cross sectional and time series data with suitable example.
2. Define multicollinearity briefly.
3. Describe BLUE properties briefly.
4. How do you measure the goodness of fit in multiple regression analysis. Discuss two model selection criteria briefly.
5. Suppose VIF for three independent variables X1, X2, and X3 are 2.1, 14.2, and 1.18 respectively then what decision do you make for the multicollinearity?

Group B **[5×6= 30]**

6. Explain assumptions of multiple regression model and BLUE properties.

OR

The following incomplete ANOVA table is the regression output which was obtained from a study of architectural firms.
Where,
The dependent variable is the total amount of fees in millions of dollars and independent variables are X_1 number of architects employed by the company (X1), number of engineers employed by the company (X2), number of years involved with health care projects (X3), number of states in which the firm operates (X4), and percent of the firm's work that is health care related (X5)

ANOVA TABLE

Source	Sum of Squares	DF	Mean Sum of Squares	F	p-value
Regression	3710	?	?	?	0.000
Error	?	26	?		
Total	6357	31			

a) Complete above ANOVA table.
b) Compute standard error of estimate and interpret its meaning.
c) Compute coefficient of determination and interpret its meaning.
d) Conduct overall test of hypothesis (use 0.05 significance level).

7. Show that $\hat{\beta} = (X'X)^{-1}(X'\underline{Y})$

8. A state fisheries commission wants to estimate the number of bass caught in a given lake during a season in order to restock the lake with the appropriate number of young fish. The commission could get a fairly accurate assessment of the seasonal catch by extensive "netting sweeps" of the lake before and after a season, but this technique is much too expensive to be done routinely. Therefore, the commission samples a number of lakes and records Y (the seasonal catch, thousands of bass per square mile of lake area), X1 (the number of lakeshore residences per square mile of lake area), X2 (the size of lake in square miles), X3 (X3=1, if the lake has public access, 0 if not) and X4 (a structure index. (Structures are weed beds sunken trees, drop-offs and other living places for bass.) The commissions run the data in SPSS to fit the regression model and the incomplete output is obtained as follows:

Predictor	Unstandardized Coefficient		t-value	p-value
	Coefficient	Standard error		
Constant	-1.9378	0.9081	?	
Residence (X1)	0.0193	0.0101	?	0.560
Size (X2)	0.3323	0.2458	?	0.120
Access (X3)	0.8355	0.2250	?	0.001
Structure (X4)	0.0477	0.0050	?	0.001

a) Complete above coefficient table and describe the best-fitting regression equation for these data.

b) Test the significant impact of X3 (Access) on Y (the seasonal catch, thousands of bass per square mile of lake area) at 5% level of significance.

9. State the unbiased ness property of estimator of slope/regression coefficient.

10. Explain the consequences of autocorrelation on unbisedness, bestness and consistent properties of OLS.

OR

Suppose the residuals for a set of data collected over 10 consecutive time period are as follows. Let the model consists of two independent variables

Time	1	2	3	4	5	6	7	8	9	10
Residuals	-5	-4	-3	-2	-1	1	2	3	4	5

Compute the Durbin Watson test statistic and test for significance at .level 05. Is there evidence of positive autocorrelation among residuals?

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Tribhuvan University

Institute of Science and Technology

SCHOOL OF MATHEMATICAL SCIENCES

First Re-Assessment 2083

Subject: **Regression Analysis**

Course No: **MDS 605**

Level: **MDS / II Year / III Semester**

Full marks: **45**

Pass Marks: **22.5**

Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A **[5x3=15]**

1. Describe BLUE properties briefly.
2. Describe the null and alternative hypothesis and state the test statistic and decision criteria to test the significance of individual regression coefficient (βi).
3. Explain three detection techniques of multicollinearity.
4. Suppose, n= 25, k= 3, sum of squares due to regression = 350, sum of squares due to error = 150, then compute multiple coefficient of determination and adjusted R^2.
5. Show that var(€) = $\sigma^2 I$.

Group B **[5×6= 30]**

6. Describe the general regression model and also describe the assumptions of the regression model.

OR

Show that $\underline{\hat{\beta}} = (X^t X)^{-1} (X^t \underline{Y})$

7. State the unbiased ness property of estimator of slope/regression coefficient.
8. A sample of 34 stores in a chain is selected for a test-market study of Pokhara Noodles. All the stores selected have approximately the same monthly sales volume. Measurements are obtained by taking price and promotional expenditure as independent variables and sales as a dependent variable. The partial output obtained from SPSS Software is as following.

ANOVA

Model	Sum of square	df	Mean square	F	p-Value
Regression	39472731	?	?	?	0.000
Residual	?	?	?		
Total	52093678	?			

Coefficients

Model	Unstandardized Coefficient		T	p-value
	Bi	Standard error		
Constant	5837.521	628.150	?	
Price	-53.217	6.852	?	0.000
Promotion	3.613	0.685	?	0.000

a) Test the significance effect of promotion on sales at 1% level of significance.

b) What percentage of total variation in sales is explained by price and promotional expenditure?

c) Compute adjusted coefficient of determination and interpret its meaning.

d) Compute standard error of estimate and interpret its meaning.

e) Test the overall fit of the model at 5% level of significance.

f) Develop the regression model and predict the sales when price Rs.70 and promotional expenditure is Rs.300.

9. Describe Durbin Watson test with null and alternative hypothesis, decision criteria and limitations.

10. Describe Breusch-Godfrey Lagrange Multiplier Test (LM test) to test the autocorrelation.

OR

Explain the consequences of autocorrelation.

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Tribhuvan University
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Second Assessment 2083

Subject: **Regression Analysis**
Course No: **MDS 605**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

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Attempt All Questions.

Group A [5x3=15]

1. show that $\underline{\hat{\beta}} = (X^t X)^{-1} (X^t \underline{Y})$
2. Explain R^2 and Adjusted R^2 with examples.
3. Explain dummy variable. If a regression equation consists of following information $\hat{Y} = 50000 - 3000X$, where Y = Salary and X = Gender (Female =1 and Male =0). Interpret the meaning of Y-intercept and slope. What are the expected salaries of male and female?
4. Explain techniques for checking normality.
5. Describe forward selection procedure briefly.

Group B [5×6= 30]

6. Explain assumptions of multiple regression model and BLUE properties.

OR

Multiple regressions is used by accountants in cost analysis to shed light on the factors that cause costs to be incurred and the magnitudes of their effects. The independent variables of such a regression model are the factors believed to be related to cost, the dependent variable. In some instances, however, it is desirable to use physical units instead of cost as the dependent variable in a cost analysis. This would be the case if most of the cost associated with the activity of interest is a function of some physical unit, such as hours of labor. The advantage of this approach is that the regression model will provide estimates of the number of labor hours required under different circumstances and these hours can then be costed at the current labor rate (Horngren, Foster. and Datar, *1994)*. The sample data have been collected from a firm's accounting and production records to provide cost information about the firm's shipping department.

The SPSS computer printout for fitting the model $Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + e$ is provided as follows:

Coefficient table:

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X_1	2.73	2.27	?	0.125
X_2	0.05	0.09	?	0.231
X_3	-2.58	0.64	?	0.001

Where Y = Labor (in Hrs)
X_1= Pounds Shipped (in 1000)
X_2= Percentage of Units Shipped by Truck
X_3= Average Shipment Weight

a) Complete the above coefficient table.
b) What is the best-fitting regression equation for these data? What number of labor hours would you expect for 33,000 pounds shipped, 90 percentages of units shipped by truck and 20 average shipment weights.
c) Test the significance of average shipment weight on number of labor hours required. X_2

7. Explain the impact of multicollinearity on BLUE properties. Also describe VIF method to detect the multicollinearity.

8. Describe Breusch-Godfrey Lagrange Multiplier Test (LM test) to test the autocorrelation.

OR

Suppose the residuals for a set of data collected over 10 consecutive time period are as follows. Let the model consists of two independent variables

Time	1	2	3	4	5	6	7	8	9	10
Residuals	-5	-4	-3	-2	-1	1	2	3	4	5

Compute the Durbin Watson test statistic and test for significance at .level 05. Is there evidence of positive autocorrelation among residuals?

9. State the sources of heteroskedasticity. Explain the Goldfeld-Quandt test to test the heteroskedasticity.

10. Explain logit model and techniques of measuring the good ness of fit in logistic regression briefly.

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$d = \Sigma e_t - \Sigma(e_{t-1})$

$d = \Sigma e_t - (e_{t-1})$

$\Sigma e_t -$

$e_t - e$

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Second Reassessment 2083

Subject: **Regression Analysis**
Course No: **MDS 605**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. Explain BLUE properties briefly.

2. The following regression output is obtained from R

Variable	Estimate	Std. Error	t-value	p-value
Intercept	15.32	2.18	7.03	<0.001
Income	0.48	0.09	5.33	<0.001
Age	-0.12	0.05	-2.40	0.020
Education	1.10	0.37	2.97	0.005

a) Write the fitted regression equation.
b) Which predictors are statistically significant?
c) Interpret the coefficient of Income and Education.

3. The software provides the following output:

Variable	VIF
X_1	2.45
X_2	11.62
X_3	3.28

1-5
5-10
10 p0.

Which variable shows evidence of multicollinearity? Suggest technique to reduce multicollinearity.

4. A researcher estimates the following model to examine the effect of education and gender on annual income:

$Income = 120000 + 18000(Education) + 25000(Male)$

where

Education = Years of education

Male = 1 if male, 0 if female

a) Interpret the coefficient of the dummy variable.
b) Predict the annual income of:

A male with 16 years of education.

A female with 16 years of education.

c) What is the expected income difference between a male and a female with the same education level?

5. Differentiate among Forward Selection, Backward Elimination, and Stepwise Regression.

Group B [5×6= 30]

6. Explain model specification and assumptions of multiple regression analysis.

7. Explain the consequences of autocorrelation on the assumptions and properties of the Ordinary Least Squares (OLS) estimators. Discuss the remedial measures that can be adopted to correct the problem of autocorrelation in econometric models.

Or

A researcher estimates a consumption function using annual data from 2011–2025 (15 years). The following output is obtained:

$R^2 = 0.93$

$F = 105.64$ ($p < 0.001$)

Durbin–Watson = 0.71

Breusch–Godfrey test p-value = 0.0003

a. Explain whether the model suffers from autocorrelation.
b. Compare the conclusions from the Durbin–Watson and Breusch–Godfrey tests.
c. Explain how autocorrelation affects the efficiency of OLS estimators.
d. Describe two econometric methods that could be used to correct the problem.

8. Define multicollinearity and distinguish between perfect and imperfect multicollinearity. Describe the methods used to detect multicollinearity.

9. Discuss the methods used to detect heteroskedasticity in a regression model.

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Describe Goldfeld -Quandt Test to test the heteroskedasticity briefly.

10. Define logistic regression with its assumptions. Also explain Hosmer Lemeshow test and ROC curve.

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